Fuel cycle needs of deploying advanced reactors

UTK NE Department Colloquium

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University of Illinois at Urbana-Champaign

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ILLINOIS

Outline

1. About Me
2. Introduction
   - Motivation
   - Background
   - Objectives
3. Transition analysis
   - Methodology
   - Results
4. Sensitivity analysis
   - Methodology
   - Results
5. Conclusions

My background

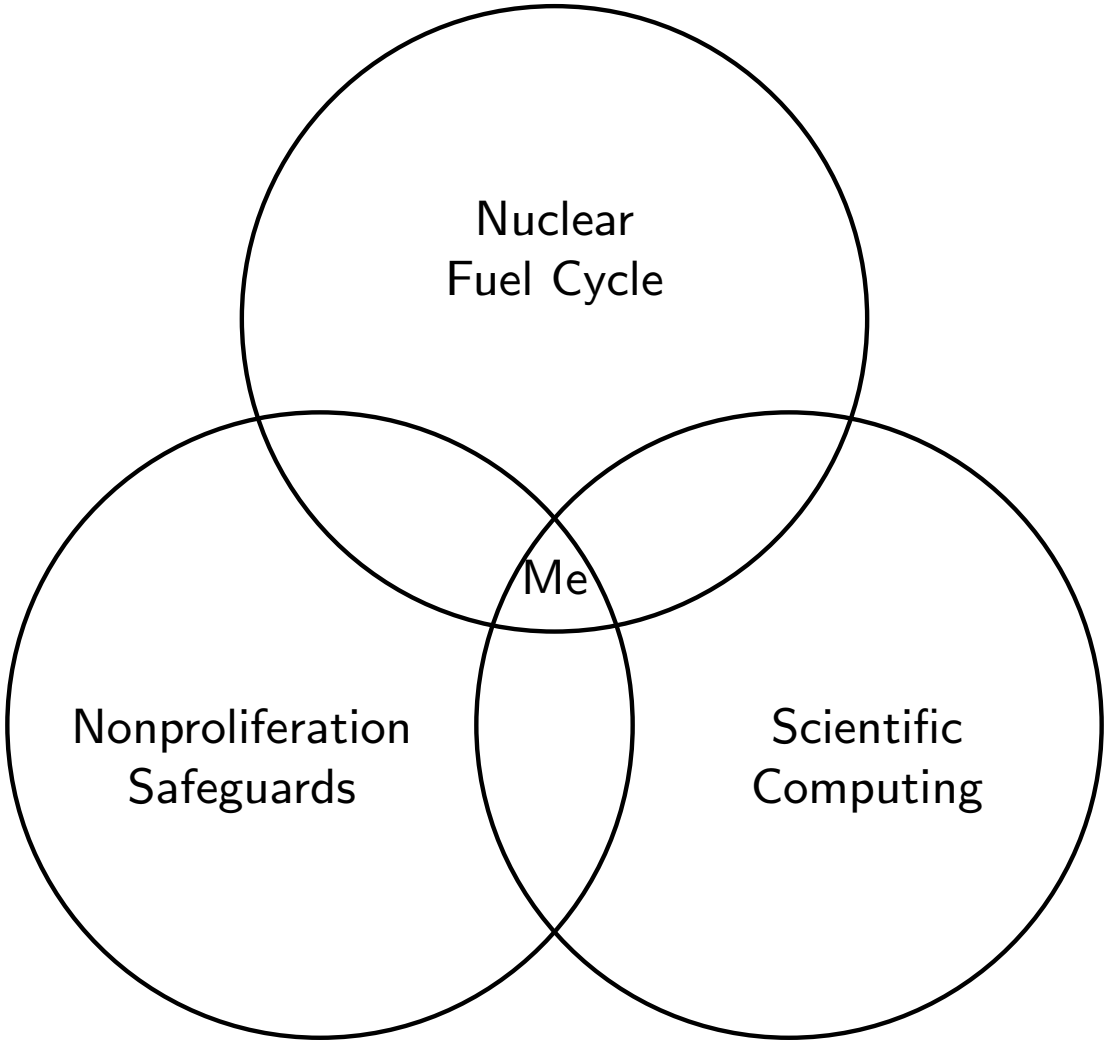
Education

- BS in Nuclear Engineering, University of Tennessee, Knoxville (2019)
- MS in Nuclear Engineering, University of Tennessee, Knoxville (2020)
- PhD in NPRE, University of Illinois Urbana-Champaign (In Progress, est. 2023)

Research Experience

- Multivariate modeling of radiation signatures for safeguards, using ORIGEN
- Modeling material flow through a pyroprocessing facility, using ORIGAMI
- Comparing effects of Doppler broadening methods in SHIFT (ORNL)
- Investigating fuel cycle impacts of using High Assay Low Enriched Uranium (HALEU) in reactors, using Cyclus, Dakota, and OpenMC (UIUC/ANL)

Research Interests



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The US is looking to develop supplies of HALEU

- Multiple new reactor designs require HALEU fuel, which allows for:
  - Longer cycle time
  - Increased capacity factor
  - Higher burnup

Table 1: Categories of uranium enrichment by weight fraction of ^{235}U.

Category	Weight fraction (%)
Depleted	$<$0.711
Natural	0.711
LEU	0.711-20
HALEU	5-20
HEU	$\geq$20

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  - Longer cycle time
  - Increased capacity factor
  - Higher burnup
- The US does not have any commercial supplies of HALEU
- To meet the HALEU demand, the U.S. DOE has proposed two methods [6]:
  - Enrichment of natural uranium
  - Recovery and downblending of HEU

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Nuclear fuel cycle

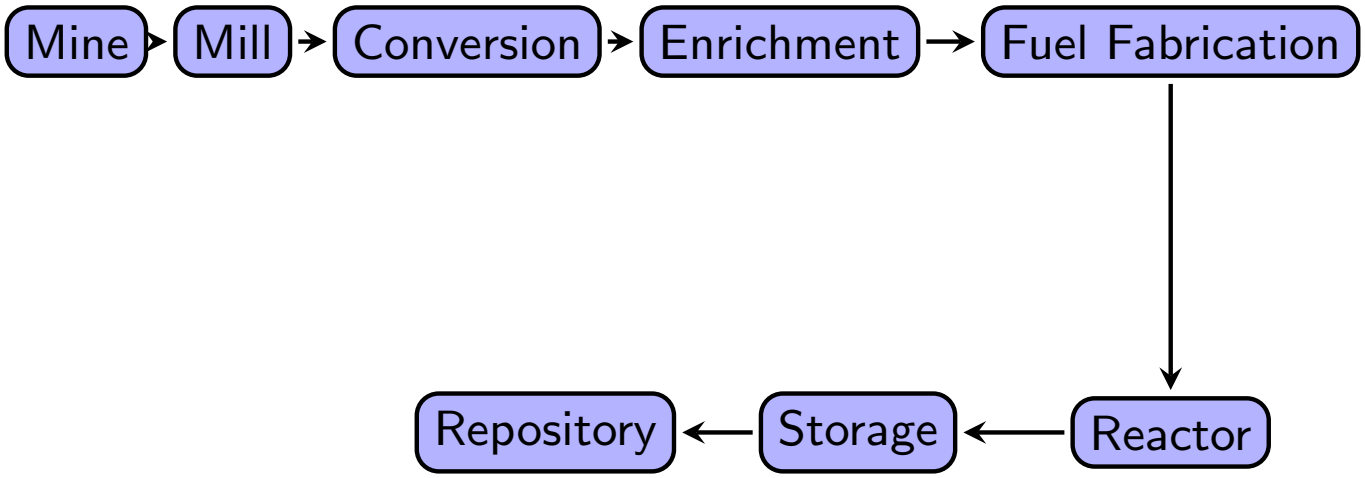
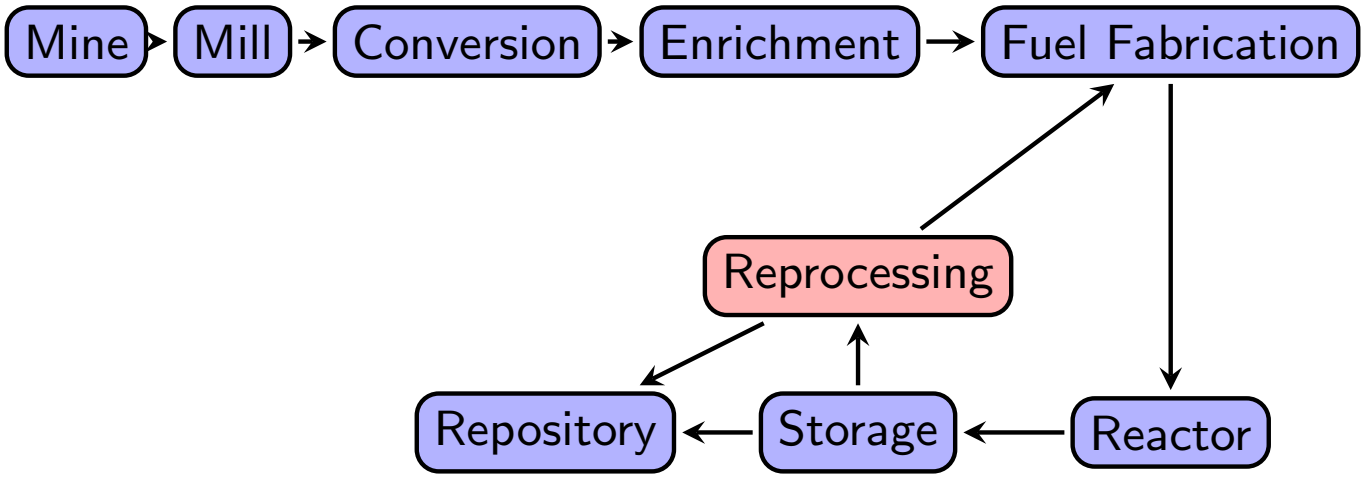


Figure 1: Overview of the Nuclear Fuel Cycle.

Nuclear fuel cycle



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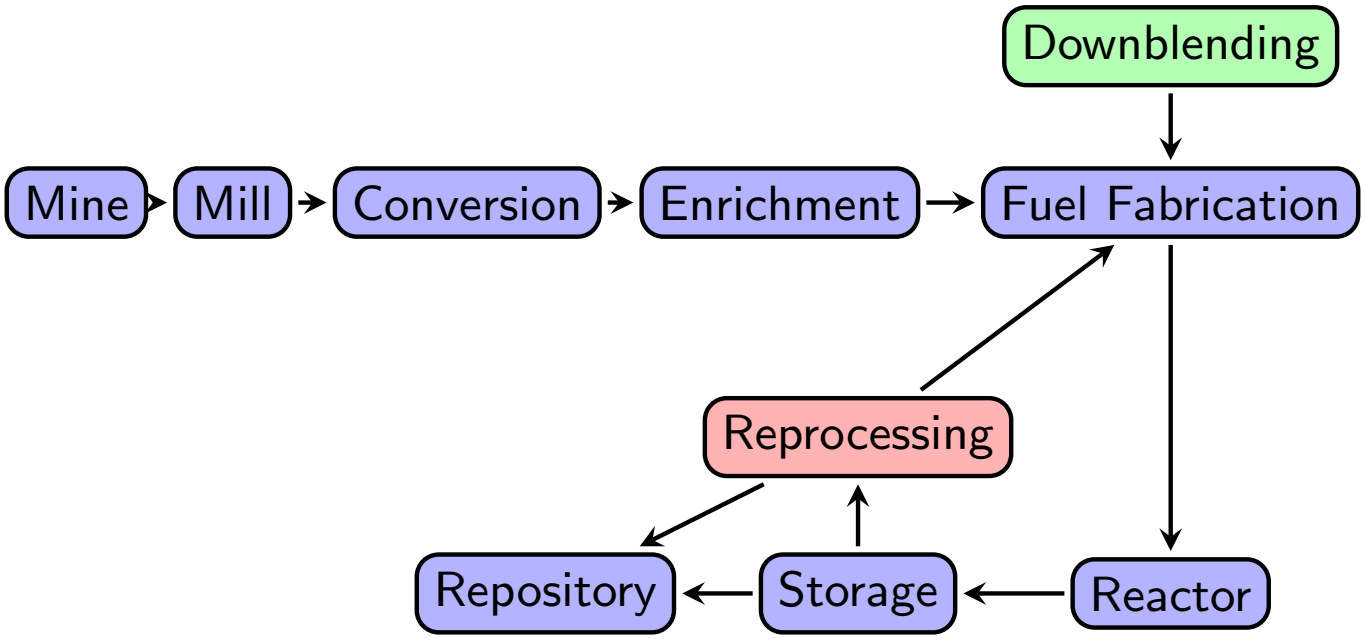

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- Process to increase the relative abundance of specific isotopes of an element

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- The throughput of a facility is based on the product mass, product assay, and the SWU capacity

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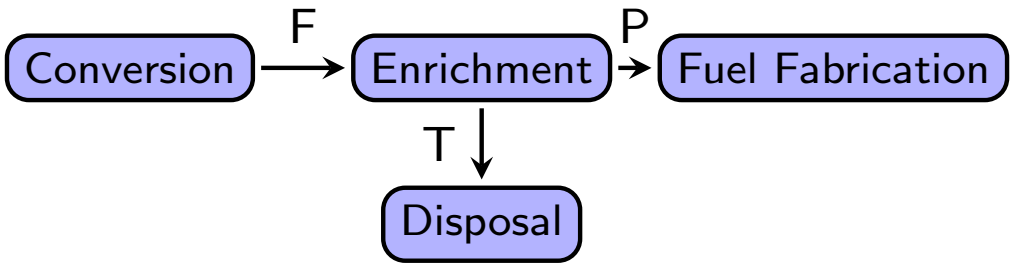
$$F = P + T$$

$$x_f F = x_p P + x_t T$$

$$SWU = [P \times V(x_p) + T \times V(x_t) - F \times V(x_f)] * t$$

in which:

$$V(x_i) = (2x_i - 1) * \ln\left(\frac{x_i}{1 - x_i}\right)$$



Variable	Definition
F	feed mass
P	product mass
T	tails mass
x_i	assay of material stream
SWU	Separative Work Units
$V(x_i)$	separation potential function

Efforts to estimate HALEU needs

Efforts are underway to estimate potential HALEU needs:

- Nuclear Energy Institute (NEI) surveyed multiple reactor design companies to estimate HALEU needs between now and 2035 [8, 10]
- DOE labs modeled the transition to some HALEU-fueled reactors to estimate HALEU needs to meet current net-zero carbon goals in 2050 [5]

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This previous work is all based on announced advanced reactor projects.

- What happens if we scale up on reactor deployment really quickly?
- How does decision making affect the fuel cycle requirements?

Objectives

- Explore how the deployment of HALEU-fueled reactors affects the US nuclear fuel cycle
- Quantify potential material requirements for the transition from Light Water Reactors (LWRs) to advanced reactors in a once-through and recycling fuel cycle
- Understand the impacts of fuel cycle parameters on the material requirements and design optimized transition scenarios
- Identify potential limitations in using downblended HEU in advanced reactors

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Fuel cycle simulators

To quantify potential material needs, we can model the transition from the current LWR fleet to advanced reactors

- Use fuel cycle simulators to model the transition
- Model the deployment and decommissioning of fuel cycle facilities
- Model material transactions between facilities
- Quantify material requirements to understand potential HALEU demand

For this work, I am using Cyclus [7]

Transition analysis assumptions

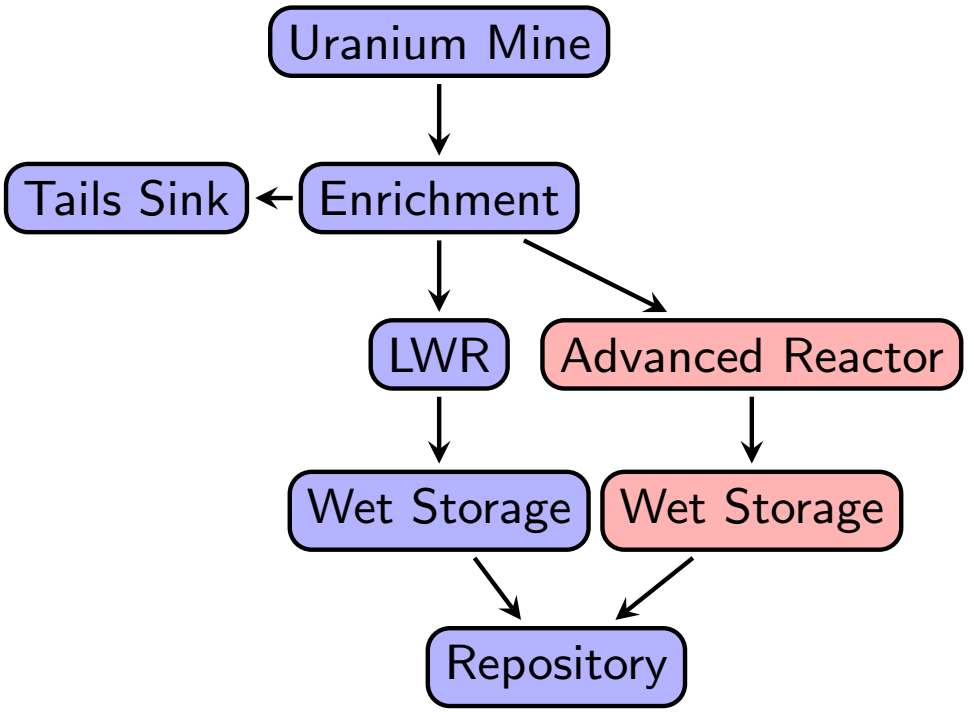

Figure 2: Fuel cycle facilities and material flow between facilities. Facilities in red are deployed at the start of the transition.

Provide fuel cycle parameters to Cyclus

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- Transitions begin in 2025

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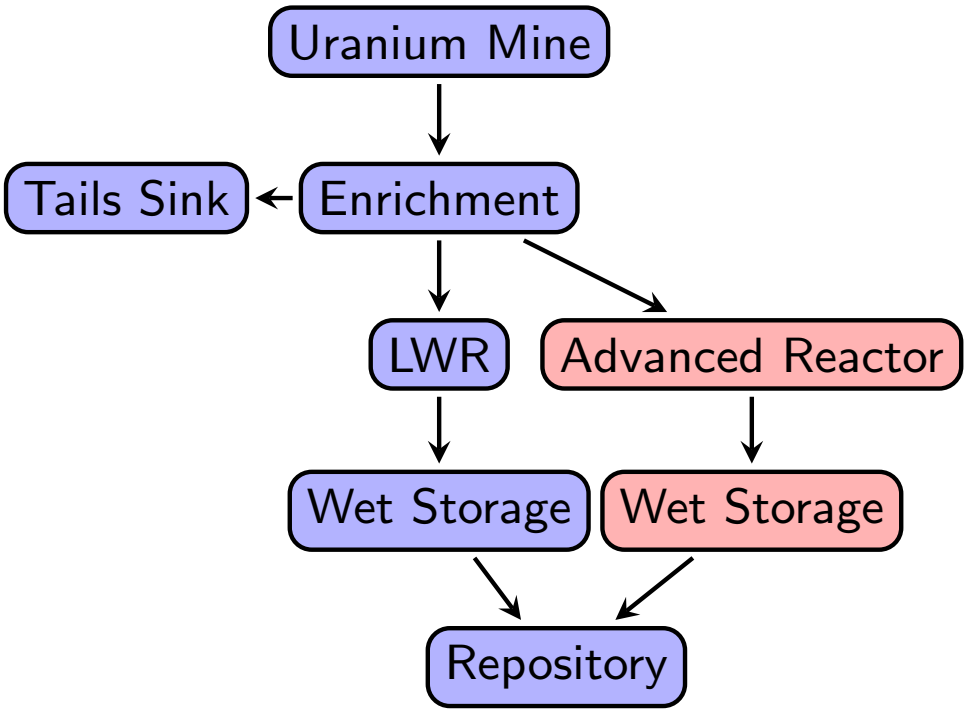

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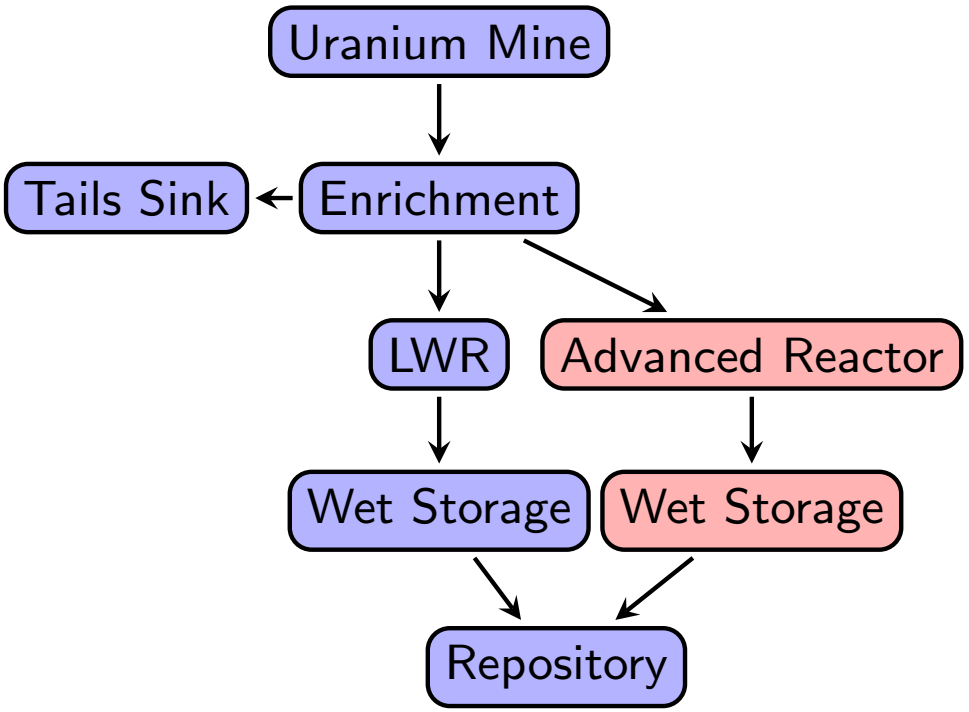

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- LWR commission dates are obtained from the IAEA PRIS database [1]
- LWRs are assumed to operate until their current license expires
- Manually calculate advanced reactor deployment
- Assume natural uranium is enriched to produce all fuel

Advanced reactors

Table 2: Advanced reactor design specifications

Design Criteria	USNC MMR [2]	X-energy Xe-100 [9]	NuScale VOYGR [11, 12, 13]
Reactor type	HTGR	HTGR	SMR
Fuel type	UO_2 FCM	UCO TRISO	UO_2 pellets
Power (MWe)	5	80	77
Power (MWth)	15	200	250
Enrichment (% ^{235}U)	19.75	15.5	4.09
Cycle Length (yr)	20	Online	1.5
Number of cycles	1	6	3
Reactor Lifetime (yr)	20	60	60
Burnup ($\frac{MWd}{kgU}$)	82	168	45

$$\text{mass (kg)} = \frac{\text{Power (MWth) * cycle length (d)*number of cycles}}{\text{Burnup (MWd/kg)}} \quad (1)$$

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Once-through scenario definitions

Table 3: Summary of the once-through fuel cycle transition scenarios. Energy growth is relative to energy from LWRs in 2025.

Scenario number	Reactors present	Energy demand
1	LWRs	N/A
2	LWRs and MMR	No growth
3	LWRs and Xe-100	No growth
4	LWRs, Xe-100, and MMR	No growth
5	LWRs, MMR, and VOYGR	No growth
6	LWRs, Xe-100, and VOYGR	No growth
7	LWRs, Xe-100, MMR, and VOYGR	No growth
8	LWRs and MMR	1% growth
9	LWRs and Xe-100	1% growth
10	LWRs, Xe-100, and MMR	1% growth
11	LWRs, MMR, and VOYGR	1% growth
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Advanced reactor deployment scheme

Manually calculate the deployment scheme for advanced reactors

- Preferentially deploy reactors with larger power output first
- Deploy reactor with largest power output until an oversupply of power would be produced, deploy the next reactor until an oversupply of power, then deploy the last reactor until demand is met
- Deployment schedule is given to Cyclus

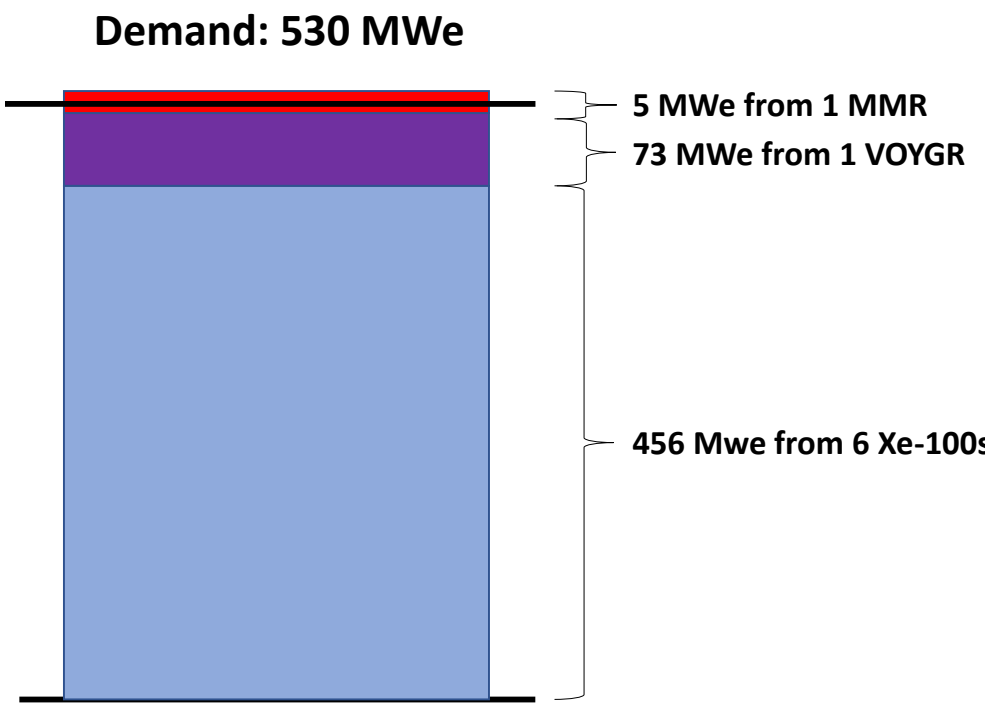

Figure 3: Example of how advanced reactors in Scenario 7 are deployed to meet a demand of 530 MWe.

Reactor design drives the uranium mass required

- Scenario 5 (MMR + VOYGR) requires the largest average mass of enriched uranium
- Scenario 2 (MMR) requires the largest mass of HALEU
- Scenario 3 (Xe-100) requires the smallest mass of enriched uranium
- Scenario 5 (MMR + VOYGR) requires the smallest mass of HALEU

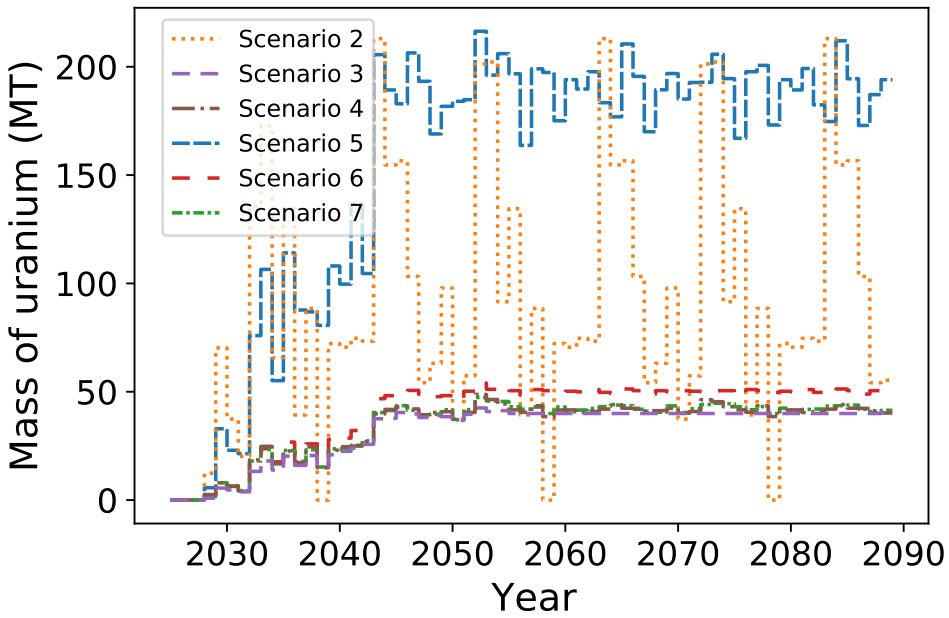

Figure 4: Annual average mass of enriched uranium required to fuel advanced reactors in Scenarios 2-7.

SWU capacity is a function of product mass and assay

- Scenario 2 (MMR) requires the largest average SWU
- The other scenarios are comparable for the average capacity they require
- This result does not capture cascade configurations or facility categories

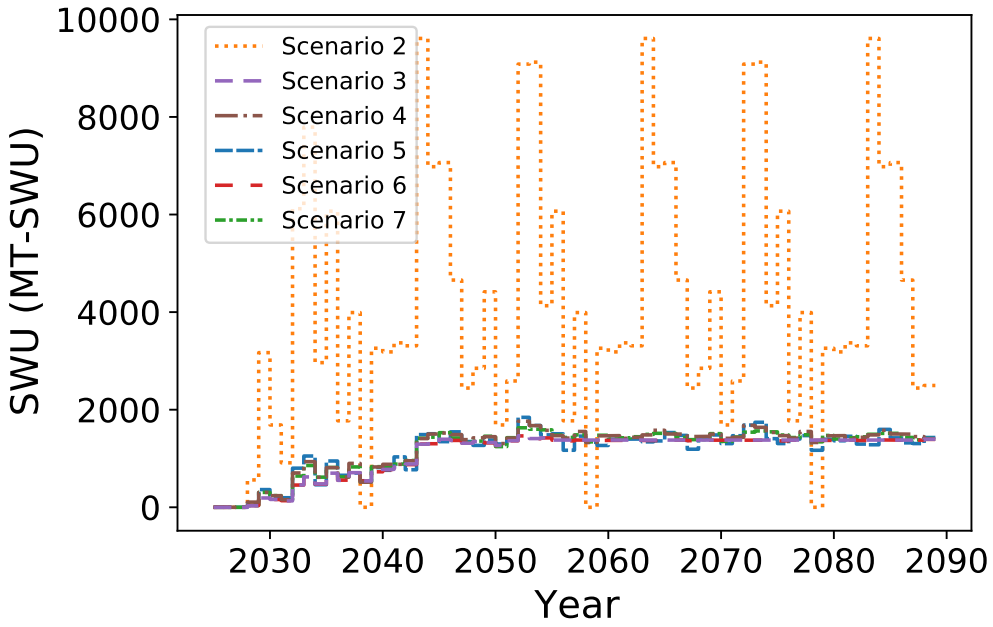

Figure 5: Annual average SWU capacity required to produce enriched uranium for and advanced reactors in Scenarios 2-7.

SNF discharged follows with fuel mass

- Scenario 5 (MMR + VOYGR) discharges the largest mass of spent nuclear fuel (SNF)
- Scenario 3 (Xe-100) discharges the least SNF
- Scenario 2 (MMR) discharges SNF the latest
- Cumulative masses range between 25,654 MT (Scenario 3) to 112,913 MT (Scenario 5)

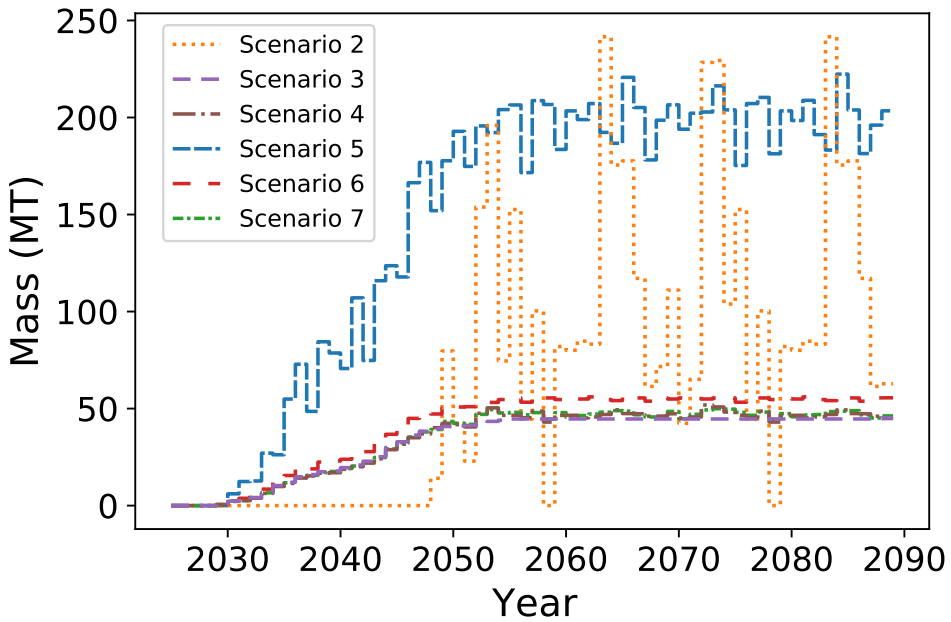

Figure 6: Annual average mass of SNF discharged from advanced reactors in Scenarios 2-7.

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Transition analysis doesn't tell the full story

- From the transition analysis we learned how changing which reactor is predominantly built affects the material requirements
- There are so many more parameters and assumptions in the scenario modeled
- We can use sensitivity analysis to see the effect of other model parameters on the material requirements

Sensitivity analysis provides insight into how decisions affect material needs

- Vary a single parameter at a time and observe the output
- Helps to identify different relationships about the transition
- Couple Cyclus with Dakota [3], model variations in Scenario 7

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Model inputs (parameters)

- Transition start time, January 2025-January 2040
- LWR lifetime: percent of LWR fleet operating for 80 years, 0-50%
- Build share of each advanced reactor, 0-50%
- MMR & Xe-100 discharge burnup

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Model outputs (metrics, material requirements)

- Total fuel mass
- HALEU mass
- Total SWU
- SWU to produce HALEU
- Feed to produce HALEU
- SNF mass

Adjust deployment when varying the build share

- Deploy the specified reactor first to meet the build share
- Deploy the other reactors using the previous scheme (largest to smallest power output) to meet the remaining demand
- Deployment schedule is given to Cyclus

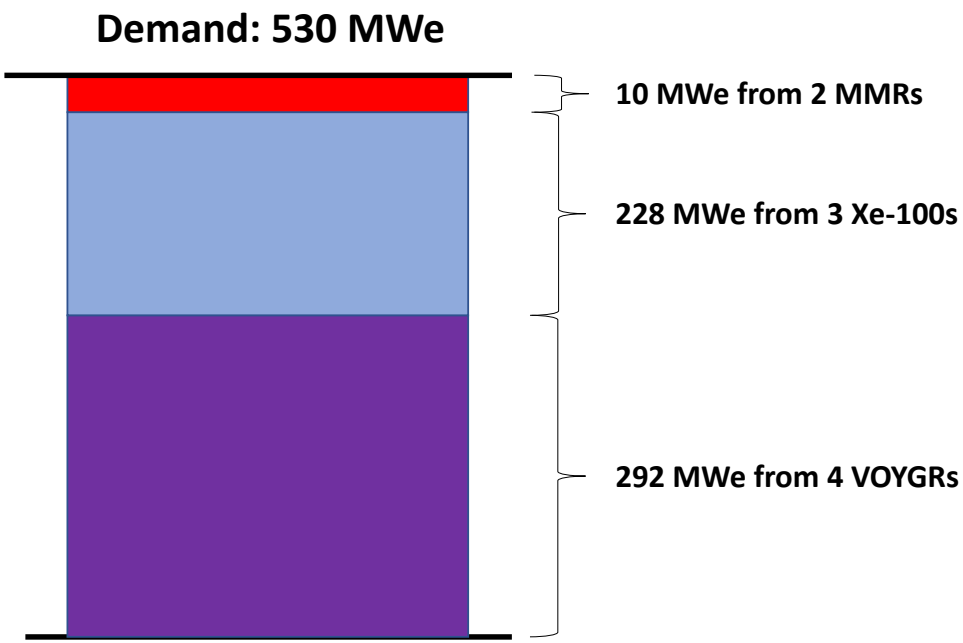


Figure 7: Deployment of advanced reactors to meet a demand of 530 MWe and a 50% VOYGR build share.

Varying input: MMR build share

- All of the metrics increase as build share increases
- SNF mass has the smallest increase in relative change
- HALEU SWU capacity has the largest relative increase
- These results are primarily because of the deployment scheme

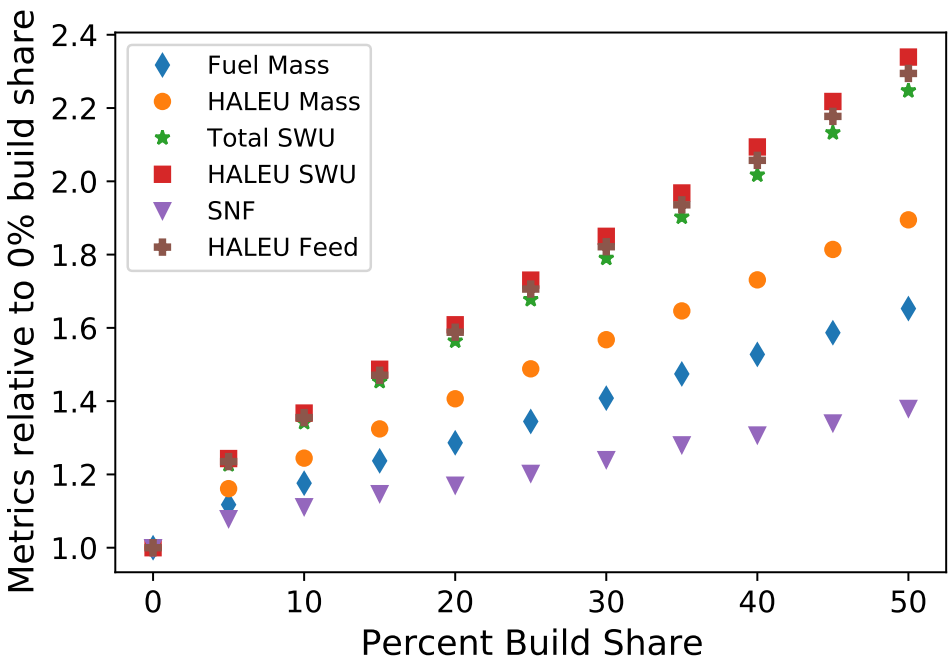
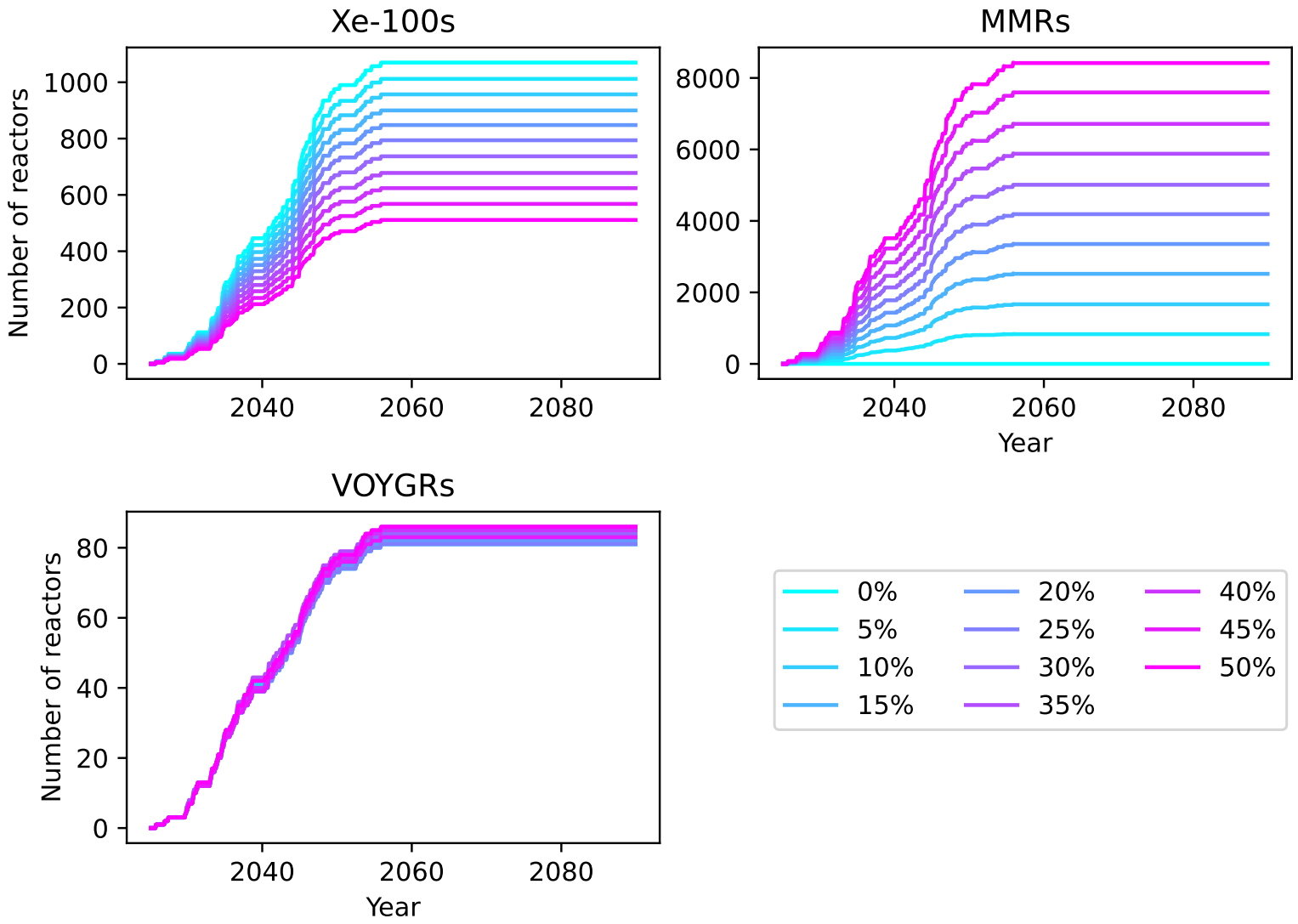


Figure 8: Relative change in metrics as a function MMR build share.

Varying MMR build share – Number of reactors deployed



Varying input: Xe-100 build share

- The HALEU-related metrics increase with increasing build share
- Total SNF and total fuel mass decrease with increasing build share
- Total SWU capacity is relatively constant, at most a relative change of 1.06

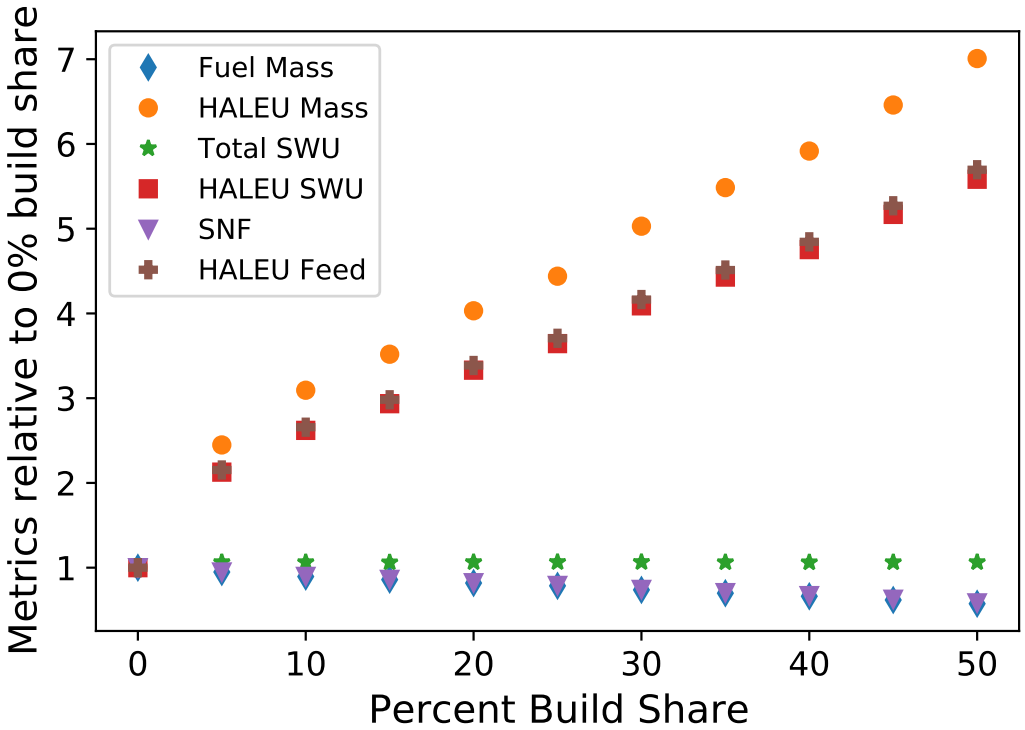


Figure 10: Relative change in metrics as a function of Xe-100 build share [4].

Varying input: VOYGR build share

- Fuel mass and SNF mass increase while HALEU metrics decrease
- Total SWU capacity is relatively constant
- Opposite trends as increasing Xe-100 build share
- As the build share increases, the number of Xe-100s decreases and the number of MMRs is fairly constant

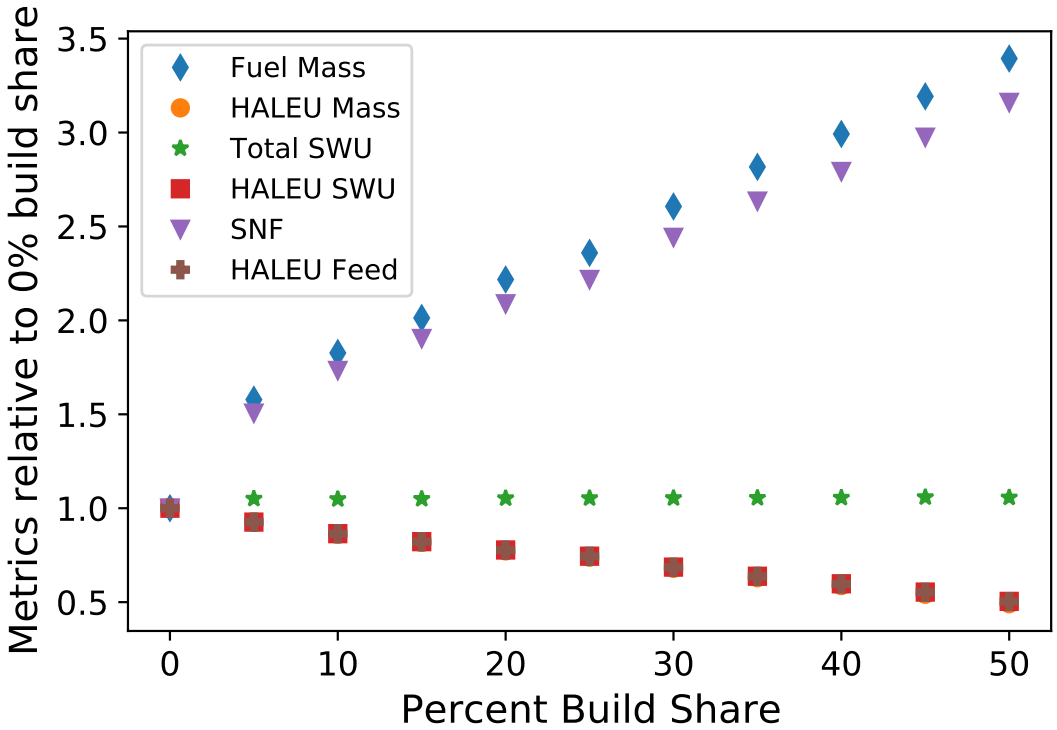


Figure 11: Relative change in metrics as a function of VOYGR build share [4].

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Conclusions

- Transition analysis and sensitivity analysis are methods to investigate the potential needs of a nuclear fuel cycle
- The reactors deployed influence the materials needed by the fuel cycle
- The deployment scheme strongly affects the material requirements
- Tradeoff between HALEU requirements and SNF discharged between the Xe-100 and VOYGR
- MMRs require more SWU capacity, but less fuel mass compared with VOYGRs
- All of the sensitivity analysis scenarios require a cumulative HALEU mass between 2,710 - 37,500 MT between 2025-2090

Future Work

Ongoing Work

- Consider recycling scenarios (close the fuel cycle)
  - Couple Cyclus with OpenMC's depletion solver
- Perform optimization of these fuel cycles to minimize material requirements
- Investigate some of the effects of using downblended uranium in advanced reactors

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Future Work

- Convert these mass/capacity needs to facility sizes and numbers
- Apply limits on uranium sources and facility throughputs
- Investigate costs for developing facilities
- Investigate safeguards for advanced reactor fuel cycles
- Research fast reactors, their fuel cycle, and safeguards for them

Acknowledgements

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- Dr. Bo Feng
- Dr. TK Kim
- ARFC Group members
- RFCA Group members

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Questions?

Enriched uranium masses

Table 4: Metrics for enriched uranium sent to advanced reactors between 2025-2090 in Scenarios 2-7.

Scenario	Average (MT/month)	HALEU Average (MT/month)	Maximum (MT)	Cumulative (MT)
2	88.90	88.90	1,007	69,251
3	31.64	31.64	86.79	24,646
4	33.62	33.62	101.5	26,193
5	152.3	3.070	581.1	118,608
6	39.92	29.51	103.3	31,095
7	33.85	32.97	103.0	26,370

SWU capacity

Table 5: Metrics for SWU capacity to enrich uranium for advanced reactors between 2025-2090 in Scenarios 2-7.

Scenario	Average (MT-SWU/month)	Average for HALEU (MT-SWU/month)	Maximum (MT-SWU)
2	4,010	4,010	45,424
3	1,090	1,090	2,991
4	1,192	1,192	3,668
5	1,145	138.5	4,228
6	1,087	1,017	3,050
7	1,167	1,161	3,735

SNF discharged

Table 6: Metrics for waste discharged from advanced reactors between 2025-2090 in Scenarios 2-7.

Scenario	Average (MT/month)	Average of HALEU (MT/month)	Maximum (MT)	Cumulative (MT)
2	66.15	66.15	1,142	51,531
3	32.93	32.93	55.29	25,654
4	34.07	34.07	77.55	26,543
5	144.9	2.294	377.0	112,913
6	40.68	30.72	75.12	31,691
7	34.46	33.62	83.14	26,848

Transition Start Time

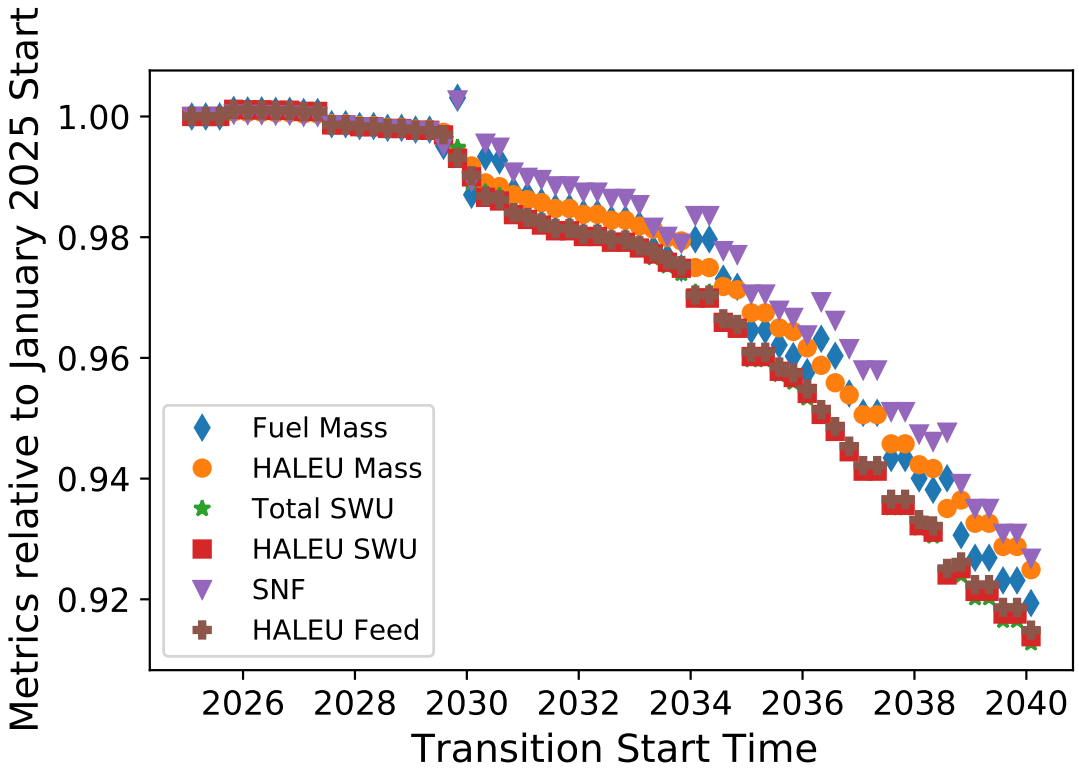

Figure 12: Change in metrics from varying the transition start time.

LWR Lifetime

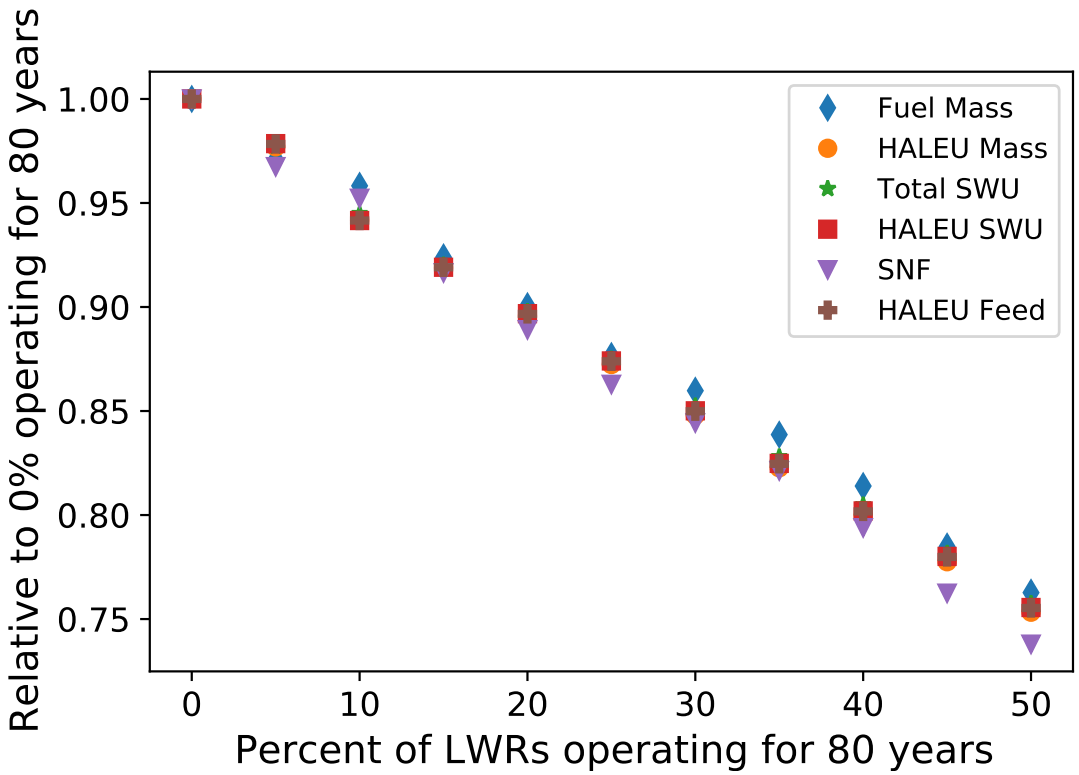

Figure 13: Change in metrics from varying the percent of the LWR fleet operating for 80 years.

MMR Burnup

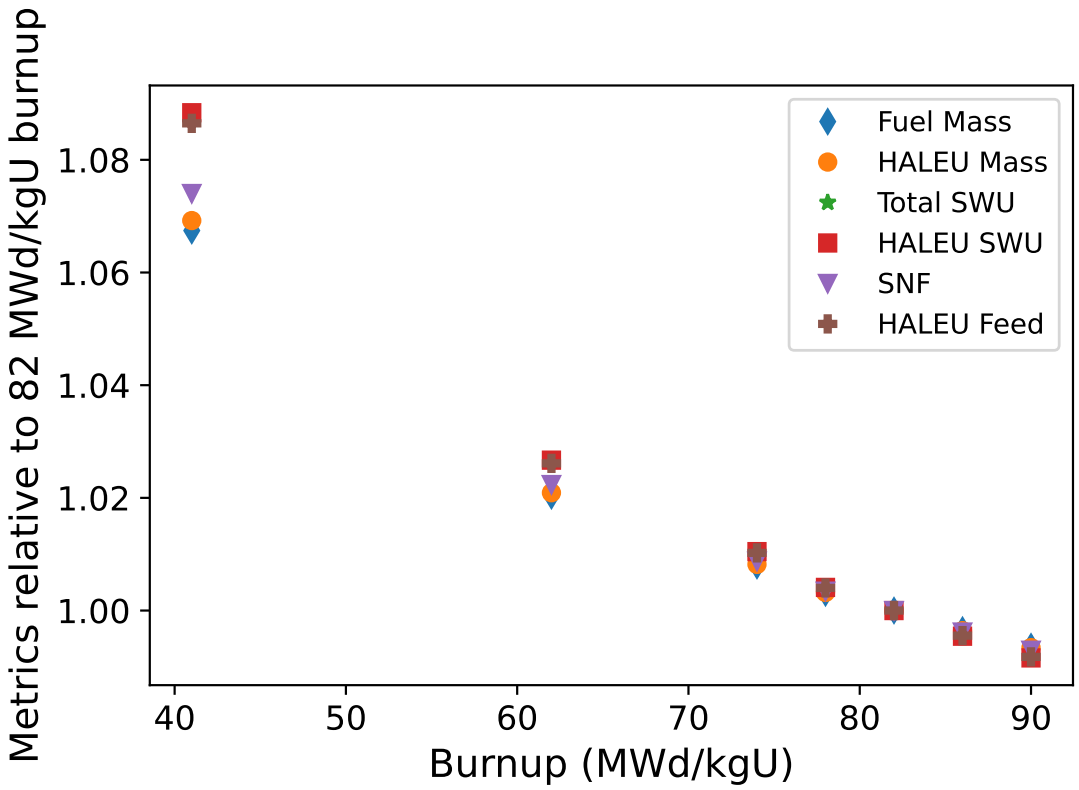

Figure 14: Change in metrics from varying the MMR burnup.

Xe-100 burnup

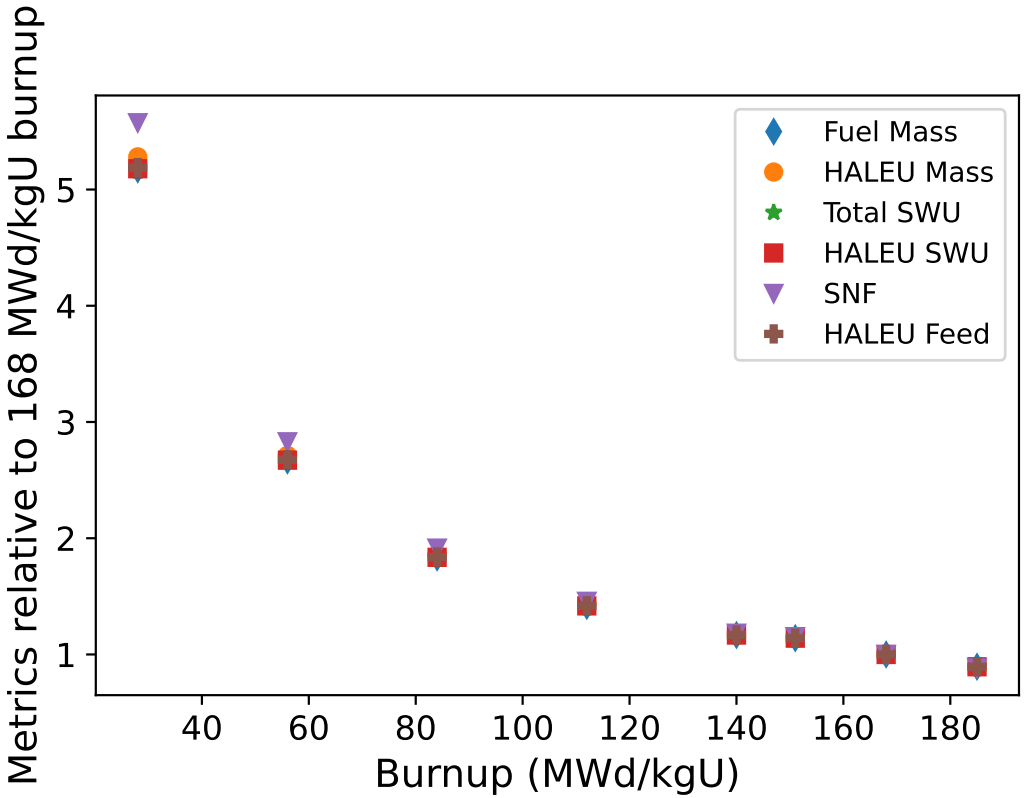

Figure 15: Change in metrics from varying the Xe-100 burnup.